

Module 03: Perception — Bayesian Inference and Signal Detection

Learning Objectives

1. Apply **Bayes' theorem** to update a belief given new evidence.
2. Explain **signal detection theory** (hits, misses, false alarms, correct rejections).
3. Connect Bayesian inference to the Active Inference concept of perception as hypothesis testing.

Introduction

Perception is not passive. Your brain does not just record what your eyes see — it actively *interprets* incoming data by combining it with prior expectations. This module introduces the mathematical framework behind this process: **Bayesian inference**.

Key Concepts

1. Bayes' Theorem

$$P(\text{hypothesis} \mid \text{data}) = \frac{P(\text{data} \mid \text{hypothesis}) \cdot P(\text{hypothesis})}{P(\text{data})}$$

In plain English: your **posterior belief** (after seeing evidence) equals the likelihood of the data times your **prior belief**, divided by a normalizing constant. This is the mathematical engine of perception.

Example: You hear a loud noise at night. Is it a burglar or the cat? Your prior says "cat is much more likely" (you have a cat, burglars are rare). The likelihood depends on the evidence: the noise sounded like a meow. Bayes' theorem combines these to give you a posterior belief: "almost certainly the cat."

2. Signal Detection Theory

In any noisy environment, an agent can make four outcomes:

	Signal Present	Signal Absent
Agent says "yes"	Hit 	False Alarm 
Agent says "no"	Miss 	Correct Rejection 

The agent's **criterion** (how much evidence it needs to say "yes") reflects its prior beliefs and the costs of errors. A smoke detector has a low criterion (better safe than sorry = many false alarms). A jury has a high criterion (beyond reasonable doubt = few false convictions).

3. Perception as Active Hypothesis Testing

In Active Inference, perception is Bayesian inference applied to the brain's generative model. The brain generates a **prediction** of what the senses should report. The incoming sensory data is the **evidence**. Bayes' theorem combines prediction and evidence to produce the brain's best guess — the **percept**. When the prediction is strong and the evidence is weak (e.g., in the dark), the prior dominates and you "see" what you expected. When the evidence is overwhelming, the prior is overridden.

Applications

- **Medical Testing:** A medical test has a 99% sensitivity (true positive rate) and 1% false positive rate. If a disease affects 0.1% of the population, what is the probability you actually have the disease given a positive test? (Students often guess 99%, but Bayes' theorem shows it is only about 9%.)
- **Optical Illusions:** The Müller-Lyer illusion (two lines that look different lengths but are the same) happens because your brain's prior about perspective depth overrides the sensory evidence.

Discussion Questions

1. You receive a positive result from a lie detector test. The test is 90% accurate. Does this prove you are lying? Use Bayes' theorem.
2. Why do radiologists sometimes miss tumors on X-rays? Explain using signal detection theory.

Summary

Perception is Bayesian inference: your brain combines prior expectations with sensory evidence to produce beliefs. Signal detection theory formalizes the tradeoffs between caution and sensitivity. Together, these ideas form the mathematical backbone of Active Inference perception.

References

- McElreath, R. (2020). *Statistical Rethinking*. Chapter 2.